

Rémi PAUTRAT

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French nationality



Education

- 07/2019 - Present **Computer Vision and Geometry group
ETH Zurich, Switzerland**
PhD in Computer Science under the supervision of Marc Pollefeys
Research interests: features detection and description, visual localization, disentanglement
- 09/2017 - 04/2019 **ETH Zurich, Switzerland**
MSc in Computer Science (Visual computing track)
Focus on deep learning, robotics and computer vision
- 09/2014 - 08/2017 **École polytechnique, Palaiseau, France**
Specialized in computer vision and machine learning
- 09/2012 - 08/2014 **Lycée du Parc, Lyon, France**
Studied maths and physics in “classes préparatoires”, an intensive two-year preparation for the competitive exams of engineering schools
- 07/2012 **A-Levels in science** with High Honours

Research & work experience

- 08/2019 - 10/2019 **TrimBot, Horizon 2020 European project**
Adapted visual SLAM to challenging environments (outdoors, with light and seasonal variations)
- 09/2018 - 04/2019 **Master thesis at the Autonomous Systems Lab (ETH Zurich)**
Worked on interactive perception: improve the vision of an ABB Yumi by interacting with the environment

- 04/2018 - 06/2018 **Open source implementation of SuperPoint**
Deep learning based features detector and descriptor
<https://github.com/rpautrat/SuperPoint>
- 03/2018 - 06/2018 **Part time work at Leica Geosystems**
Detection of doors and windows in 3D from Lidar and RGB images
- 04/2017 - 07/2017 **Research internship at INRIA Nancy**
Bayesian optimization for robotic damage recovery
- 06/2016 - 08/2016 **Internship in the robotics company MetraLabs in Germany**
- 09/2015 - 05/2016 Directed a **team project** on a multi-sensors robot (including a Kalman filter, stereovision and SLAM)

Teaching

- 09/2019 - 02/2020 Head TA in the Visual Computing course at ETH Zurich (Bachelor level in Computer Science)
- 09/2015 - 07/2019 Private maths lessons on the web platform LiveMentor

Skills

- Languages**
- French:* native language
English: advanced (TOEFL iBT: 108/120)
German: good knowledge (B1 - B2)
- IT skills**
- Languages* Python, C++, Java, Matlab, OCaml, HTML, PHP
Libraries PyTorch, Tensorflow, Scikit-learn, OpenCV, ROS

Publications

- R. Pautrat, V. Larsson, M. Oswald, M. Pollefeys, *Online invariance selection for Local Feature Descriptors*, ECCV 2020
- R. Pautrat*, T. Novkovic*, F. Furrer, M. Breyer, R. Siegwart, J. Nieto, *Object Finding in Cluttered Scenes Using Interactive Perception*, ICRA 2020
- R. Pautrat, K. Chatzilygeroudis, JB Mouret, *Bayesian optimization with automatic prior selection for data-efficient direct policy search*, ICRA 2018